

ID1063: Introduction to Programming
Aug-Nov 2023
IIT-Hyderabad
Mid-Semester Exam

October 14, 2023

Answer the questions in the spaces provided. If you run out of room, ask for an answer booklet. Read the questions carefully before answering. All the best!

Name: _____

Roll No: _____

Section 1. Answer True/False ($6 \times 1 = 6$ marks)

- _____ Identifier names in C can contain letters, digits, and underscore characters only.
- _____ Having a function prototype is a must if you use a user-defined function in a program.
- _____ The `-o` command in `gcc` specifies the filename of the compilation output.
- _____ Array is a data type to store group of elements, similar or dissimilar, consecutively in memory.
- _____ A variable declared with the `static` keyword in a function retains its value across all functions of a program.
- _____ `return` statement should always mention a constant or a variable name whose value needs to be returned.

Section 2. Multiple Choice Questions ($8 \times 1 = 8$ marks)

1. Which of the following is not a valid way to declare an array in C?

- (a) `int numbers[5];`
- (b) `int[5] numbers;`
- (c) `int[] numbers = 1, 2, 3, 4, 5;`
- (d) `int numbers[] = 1, 2, 3, 4, 5;`

2. Which one of the following statements is **true**?

- (a) The stack frame corresponding to a function under execution does not contain the global variables used by the function.
- (b) Function prototype needs to list the names of the function argument variables.
- (c) Function call stack contains frames corresponding to all functions defined in the program.
- (d) Variable reusability is an important advantage of having function calls.

3. Which of the following is NOT a valid data type in C?

- (a) `string`
- (b) `int`
- (c) `float`
- (d) `double`

4. Consider the following code snippet, and state the expected output:

```
int i;
for (i = 1; i <= 5; i++) {
    if (i % 2 == 0) {
        continue;
    }
    printf("%d ", i);
}
```

- (a) 1 2 3 4 5
- (b) 1 3 5
- (c) 2 4
- (d) None of the above

5. Which of the following is a benefit of seeding the random number generator?

- (a) Getting a different random number in each call to `rand()`
- (b) Getting a different sequence of random numbers with different seeds
- (c) Faster generation of the random number
- (d) Controlling the scaling factor for random number generation

6. Which among the following operators have the highest precedence in C?

- (a) *
- (b) %
- (c) =
- (d) ++

7. Consider the following code snippet, and state the expected output:

```
int x = 5;
printf("%d",x++);
x += 3;
printf("%d",x);
```

- (a) 6,9
- (b) 5,9
- (c) 5,8
- (d) 6,8

8. Which one of the following statements is **false**?

- (a) `double` can store a value of much greater magnitude with greater precision than `float`
- (b) `scanf` returns the number of items that were read
- (c) The `default` case is optional in a `switch-case` statement.
- (d) A `for` loop statement must always contain an *initialization*, a *condition*, and an *increment/decrement/update to the loop variable* .

Section 3. Predict the outputs ($8 \times 2 = 16$ marks)

For each code snippet, assume that the appropriate header files are already included.

Add a one-line justification for your answer.

```
1. void main()
{
```

```

int i=1, j=2;
i=i%2, j=j%2;
if(i=0) {
    printf("%d", j);
}
else {
    printf("%d", i);
}
}

```

2. `int main()`

```

{
    int n;
    for(n = 7; n!=0; n--)
        printf("n = %d ", n--);
    return 0;
}

```

3. `void main()`

```

{
    int v1=10, v2=-20, negnum=0;

    if(v1*v2 < -1) {
        int negnum=1;
    }

    if (negnum == -1) {
        printf("One number is negative");
    }
    else {
        printf("Both are positive or both are negative");
    }
}

```

4. `void main()`

```

{
    int i=1;
    printf((i%2==0)?"odd":"even");
}

```

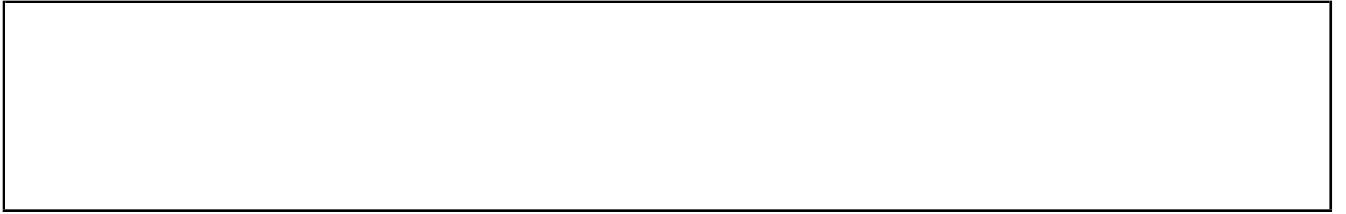
```
5. void main()
{
    int arr[3]={1};
    arr[2]=arr[0]+arr[1];
    printf("%d %d %d", arr[0], arr[1], arr[2]);
}
```

```
6. void main()
{
    int x=y=1;
    printf("%d",x-y);
}
```

```
7. void main()
{
    int i=1;
    i=i++ + ++i;
    printf("%d", i);
}
```

```
8. #include<stdio.h>
enum {ONE=1, TWO=2}

void main()
{
    printf("%d", ONE+TWO);
}
```



Answer Key for Exam A

Section 1. Answer True/False ($6 \times 1 = 6$ marks)

- True Identifier names in C can contain letters, digits, and underscore characters only.
- False Having a function prototype is a must if you use a user-defined function in a program.
- True The `-o` command in `gcc` specifies the filename of the compilation output.
- False Array is a data type to store group of elements, similar or dissimilar, consecutively in memory.
- False A variable declared with the `static` keyword in a function retains its value across all functions of a program.
- False `return` statement should always mention a constant or a variable name whose value needs to be returned.

Section 2. Multiple Choice Questions ($8 \times 1 = 8$ marks)

1. Which of the following is not a valid way to declare an array in C?
 - (a) `int numbers[5];`
 - (b) `int[5] numbers;`
 - (c) `int[] numbers = 1, 2, 3, 4, 5;`
 - (d) `int numbers[] = 1, 2, 3, 4, 5;`
2. Which one of the following statements is **true**?
 - (a) The stack frame corresponding to a function under execution does not contain the global variables used by the function.
 - (b) Function prototype needs to list the names of the function argument variables.
 - (c) Function call stack contains frames corresponding to all functions defined in the program.
 - (d) Variable reusability is an important advantage of having function calls.
3. Which of the following is NOT a valid data type in C?
 - (a) `string`
 - (b) `int`
 - (c) `float`
 - (d) `double`
4. Consider the following code snippet, and state the expected output:

```
int i;
for (i = 1; i <= 5; i++) {
    if (i % 2 == 0) {
        continue;
    }
    printf("%d ", i);
}
```

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- (b) 1 3 5
- (c) 2 4
- (d) None of the above

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- (a) `*`
- (b) `%`
- (c) `=`
- ☒ (d) `++`

7. Consider the following code snippet, and state the expected output:

```
int x = 5;
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x += 3;
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```

- (a) 6,9
- ☒ (b) 5,9
- (c) 5,8
- (d) 6,8

8. Which one of the following statements is **false**?

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Section 3. Predict the outputs ($8 \times 2 = 16$ marks)

For each code snippet, assume that the appropriate header files are already included.

Add a one-line justification for your answer.

```
1. void main()
{
    int i=1, j=2;
    i=i%2, j=j%2;
    if(i=0) {
        printf("%d", j);
    }
    else {
        printf("%d", i);
    }
}
```

Answer: 0

```
2. int main()
{
    int n;
    for(n = 7; n!=0; n--)
        printf("n = %d ", n--);
    return 0;
}
```

Answer: It will print 7 5 3 1 -1 and then go into an infinite loop

```
3. void main()
{
    int v1=10, v2=-20, negnum=0;

    if(v1*v2 < -1) {
        int negnum=1;
    }

    if (negnum == -1) {
        printf("One number is negative");
    }
    else {
        printf("Both are positive or both are negative");
    }
}
```

Answer: Both are positive or both are negative

```
4. void main()
{
    int i=1;
    printf((i%2==0)?"odd":"even");
}
```

Answer: even

```
5. void main()
{
    int arr[3]={1};
    arr[2]=arr[0]+arr[1];
    printf("%d %d %d", arr[0], arr[1], arr[2]);
}
```

Answer: 1 0 1

```
6. void main()
{
    int x=y=1;
    printf("%d",x-y);
}
```

Answer: Compile Error.

```
7. void main()
{
    int i=1;
    i=i++ + ++i;
    printf("%d", i);
}
```

Answer: 4

```
8. #include<stdio.h>
enum {ONE=1, TWO=2}

void main()
{
    printf("%d", ONE+TWO);
}
```

Answer: Compile Error. Semicolon missing after enum declaration.